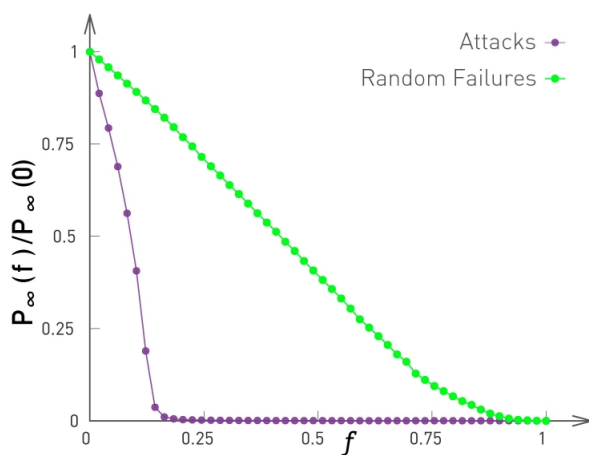


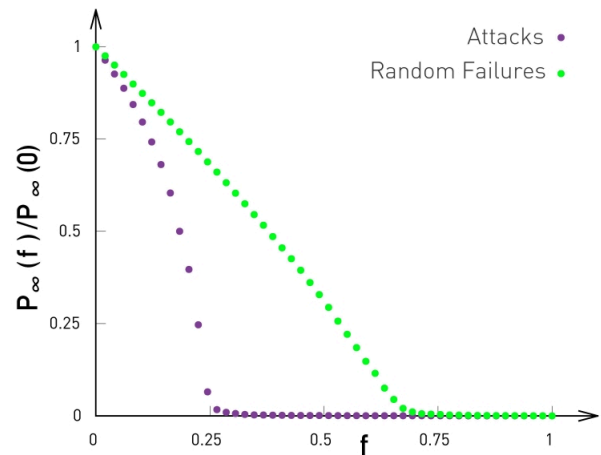
L4: Random Failures and Targeted Attacks in Power-law Networks

Another interesting characteristic of power-law networks is that they behave very differently than random ER-graphs (or, more generally, networks with Poisson degree distribution.) in the presence of node failures.

Let us first distinguish between **random failures** (where a fraction f of randomly selected nodes are removed from the network), and **targeted attacks** (where a fraction f of the nodes with the highest degree are removed from the network). In a communication network, for instance, random failures can be caused by router malfunctions, while targeted attacks may be caused by a terrorist that knows the topology of the network and disrupts the highest-connectivity routers first.



Scale-free Network Under Attack, Figure 8.11 from networksciencebook.com by Albert-László Barabási



Attacks and Failures in Random Networks. Figure 8.13 from networksciencebook.com by Albert-László Barabási

The plots in this page compare the effect of both random failures and attacks on both power-law networks (left) and ER-graphs (right). In both cases, the networks have the same size (10,000 nodes and 15,000 edges). The exponent of the power-law distribution in the network at the left is 2.5 (meaning that the variance of the degree distribution is infinite) and the average degree is 3.

The y-axis of both plots shows the fraction of nodes that belong to the largest connected component, for a given fraction f of removed nodes (more precisely, the y-axis shows the ratio $P_{\infty}(f)/P_{\infty}(0)$, where the numerator is the probability that a node belongs in the largest connected component given a fraction f of removed nodes, while the denominator is the same probability but when $f=0$).

In terms of random failures, the power-law degree network is much more robust than the random network: the largest connected component includes almost all nodes, even as f approaches 100% of the nodes. On the other hand, the random network's largest connected component disintegrates after f exceeds a critical threshold (around 0.7 in this example). This difference is mostly due to the presence of hubs in power-law networks: hubs have so many connections that they manage to keep the non-deleted nodes in the same connected component, even when f is close to 1.

The situation is very different from targeted attacks: power-law networks are very sensitive to those because an attacker would first delete the hub nodes, causing a disintegration of the largest connected component after f exceeds a critical threshold (*around 0.15 in this example*). That critical threshold is higher for Poisson networks because they do not have hub nodes.

In summary, power-law networks are more robust to random failures than Poisson networks – but they are also more sensitive to targeted attacks than Poisson networks.